(see Chapter 1, p. 10) and is essential for neural tube development. As the neural plate lengthens, its lateral edges elevate to form **neural folds**, and the depressed midregion forms the **neural groove** (Fig. 6.2). Gradually, the neural folds approach each other in the midline, where they fuse (Fig. 6.3*A*,*B*). Fusion begins in the cervical region (fifth somite) and proceeds cranially and caudally (Fig. 6.3*C*,*D*). As a result, the **neural tube** is formed. Until fusion is complete, the cephalic

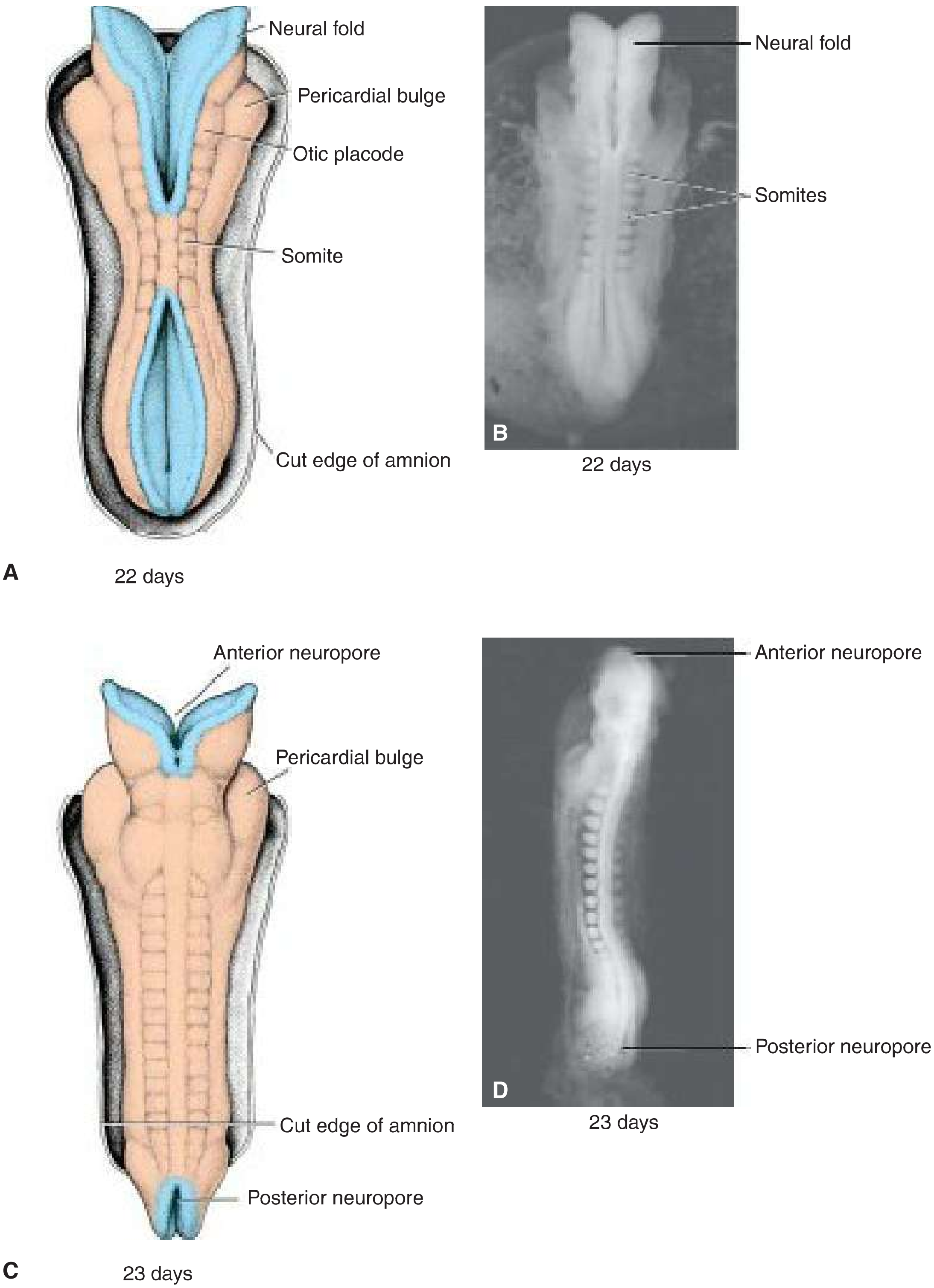

FIGURE 6.3 **A.** Dorsal view of an embryo at approximately day 22. Seven distinct somites are visible on each side of the neural tube. **B.** Dorsal view of a human embryo at 22 days. **C.** Dorsal view of an embryo at approximately day 23. Note the pericardial bulge on each side of the midline in the cephalic part of the embryo. **D.** Dorsal view of a human embryo at 23 days.

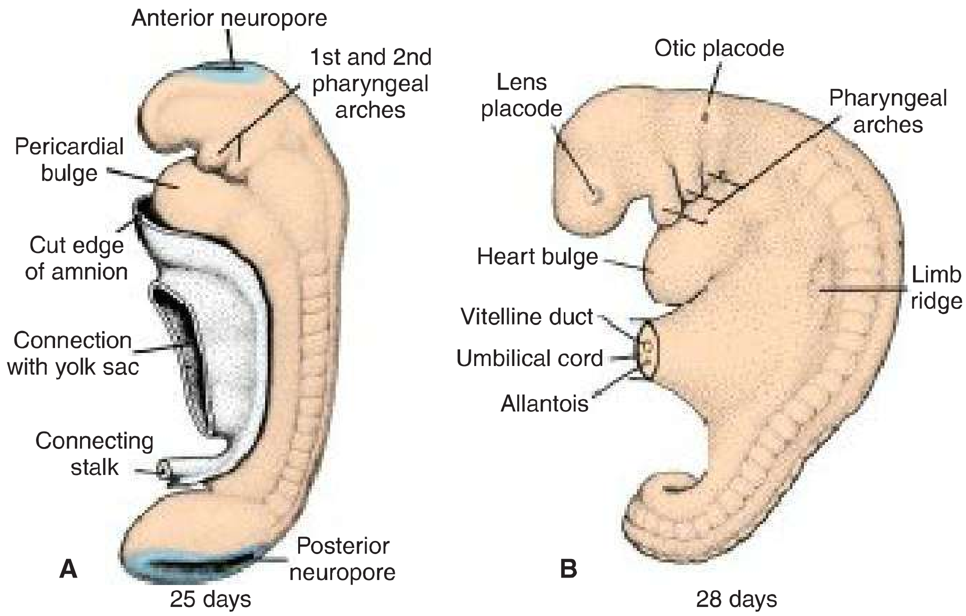

FIGURE 6.4 **A.** Lateral view of a 14-somite embryo (approximately 25 days). Note the bulging pericardial area and the first and second pharyngeal arches. **B.** The left side of a 25-somite embryo approximately 28 days old. The first three pharyngeal arches and lens and otic placodes are visible.

(cranial) and **posterior (caudal) neuropores**, respectively (Figs. 6.3*C*,*D* and 6.4*A*). Closure of the cranial neuropore occurs at approximately day 25 (18- to 20-somite stage), whereas the posterior neuropore closes at day 28 (25-somite stage) (Fig. 6.4*B*). Neurulation is then complete, and the central nervous system is represented by a closed tubular structure with a narrow caudal portion, the **spinal cord**, and a much broader cephalic portion characterized by a number of dilations, the **brain vesicles** (see Chapter 18).

Neural Crest Cells

As the neural folds elevate and fuse, cells at the lateral border or crest of the neuroectoderm begin to dissociate from their neighbors. This cell population, the **neural crest** cells (NCC; Figs. 6.5 and 6.6), undergoes an **epithelial-to-mesenchymal transition** as it leaves the neuroectoderm by active migration and displacement to enter the underlying mesoderm. (**Mesoderm** refers to cells derived from the epiblast and extraembryonic tissues. **Mesenchyme** refers to loosely organized embryonic connective tissue regardless of origin.) Crest cells from the trunk region leave the neuroectoderm **after closure** of the neural tube and migrate along one of two pathways: (1) a dorsal pathway through the dermis, where they will enter the ectoderm through holes in the basal lamina to form **melanocytes** in the skin and hair follicles, and (2) a ventral pathway through the anterior half of each somite to become **sensory ganglia**, **sympathetic** and **enteric neurons**, **Schwann cells**, and **cells of the adrenal medulla** (Fig. 6.5). NCC also form and migrate from cranial neural folds, leaving the neural tube **before closure** in this region (Fig. 6.6). These cells contribute to the **craniofacial skeleton** as well as **neurons for cranial ganglia**, **glial cells**, **melanocytes**, and other cell types (Table 6.1, p. 78). NCC are so fundamentally important and contribute to so many organs and tissues that they are sometimes referred to as the **fourth germ layer**. They are also involved in at least **one-third** of all birth defects and many cancers, such as melanomas, neuroblastomas, and others. Evolutionarily, these cells appeared at the dawn of vertebrate development and formed the basis for vertebrate features, including sensory ganglia and craniofacial structures that expanded the success of vertebrates by perfecting a predatory lifestyle.

Molecular Regulation of Neural Crest Induction

Induction of NCC requires an interaction at the junctional border of the neural plate and surface ectoderm (epidermis) (Fig. 6.5*A*). Intermediate concentrations of BMPs are established at this boundary compared to neural plate cells